Experiment 2

Title

Design a landing page for a fictional product or service using HTML, CSS, and Bootstrap. Include a hero section, product descriptions, pricing plans, and a contact form. Make it mobile friendly with Bootstrap grid system.

Theory:

A landing page is a single, focused web page designed to present a product, service, or idea to users in a clear and visually appealing manner. The primary objective of a landing page is to capture user attention, communicate value effectively, and encourage a specific action such as sign-up, inquiry, or purchase. Landing pages are widely used in modern web development for marketing, product launches, and service promotion

HTML (HyperText Markup Language)

HTML is the standard markup language used to structure web content. It defines the layout and elements of a web page such as headings, paragraphs, images, links, forms, and sections. In a landing page, HTML is used to create key components like the hero section, product descriptions, pricing tables, and contact forms. Semantic HTML elements improve accessibility and search engine optimization.

CSS (Cascading Style Sheets)

CSS is used to control the visual presentation of HTML elements. It allows developers to apply styles such as colors, fonts, spacing, alignment, and layouts. In a landing page, CSS enhances the overall look and feel, ensures consistency in design, and improves user experience. Custom CSS can be used along with Bootstrap styles to achieve a unique and professional appearance

Bootstrap Framework

Bootstrap is an open-source front-end framework that simplifies responsive web design. It provides predefined CSS classes and JavaScript components that help in building modern, mobile-first websites quickly. Bootstrap includes ready-made components such as navigation bars, buttons, cards, forms, and pricing layouts which are commonly used in landing pages.

Hero Section

The hero section is the topmost part of a landing page and serves as the first visual interaction for users. It typically contains a prominent heading, a short description, and a call-to-action button. The hero section plays a crucial role in grabbing user attention and clearly communicating the purpose of the product or service.

Product Description Section

This section provides detailed information about the product or service, highlighting its features, benefits, and unique selling points. Clear and concise descriptions help users understand the value offered and build trust. Bootstrap cards and grid layouts are often used to present product information in a structured manner.

Pricing Plans

Pricing plans display the cost structure of the product or service in a transparent and organized format. This section commonly uses columns to compare different plans side by side. Bootstrap's grid system enables easy alignment of pricing cards, making the section visually balanced and readable on all screen sizes.

Contact Form

A contact form allows users to communicate with the service provider. It typically includes input fields such as name, email, message, and a submit button. Bootstrap form classes ensure proper alignment, validation, and responsiveness, improving usability across devices.

Responsive Design Using Bootstrap Grid System

Responsive web design ensures that a web page adapts to different screen sizes such as desktops, tablets, and mobile phones. Bootstrap's grid system is based on a 12-column layout that adjusts automatically according to screen width. By using grid classes like col-sm, col-md, and col-lg, developers can create mobile-friendly layouts without writing complex CSS.

Procedure

1. Create index.html file.
2. Add Bootstrap CDN in <head>:

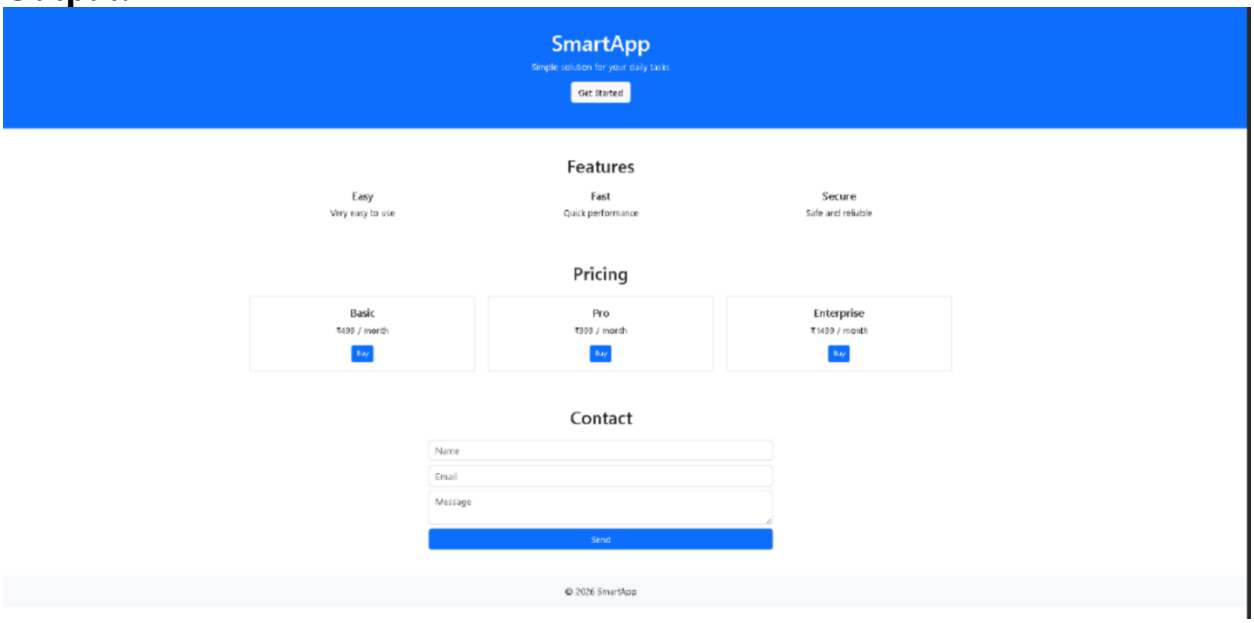
   ```
   <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
   rel="stylesheet">
   ```

3. Create hero section:

   ```
   <div class="container text-center py-5">
     <h1>My Product</h1>
     <p>Amazing solution for modern problems</p>
   </div>
   ```

4. Add product cards using Bootstrap grid.
5. Add pricing section using cards.
6. Create contact form using Bootstrap forms.

Output:



Conclusion:

Using HTML, CSS, and Bootstrap, a responsive and visually appealing landing page can be designed efficiently. Bootstrap simplifies layout management and ensures mobile friendliness, making it an ideal framework for modern web development.

Question:

1. What is a landing page?- A **landing page** is a web page that users reach after clicking a link, advertisement, or search result. It is designed to provide specific information and encourage visitors to take a particular action, such as signing up, purchasing a product, or filling out a form.

2. How do you make images and videos responsive in Bootstrap?- In Bootstrap, images and videos are made responsive by using built-in responsive classes. The class

img-fluid is used for images, which automatically adjusts the image size according to the screen size by setting the maximum width to 100% and height to auto.

For videos, responsive embedding is done using Bootstrap's **ratio classes** such as ratio and ratio-16x9, which make videos automatically resize based on the device screen.

Example:

Image: <img src="image.jpg" class="img-fluid">
Video: ...

3. What is the difference between margin and padding?- **Margin** is the space outside the border of an element. It creates distance between one element and other surrounding elements.

   Padding is the space inside the border of an element. It creates space between the content and the element's border.